new squeeze/pressure balance can be achieved—one with the star slightly smaller than before. Thus, as Sirius B continues gradually to cool by radiating heat into interstellar space, it must gradually shrink in size.

How does this gradual shrinkage end? What will be the ultimate fate of Sirius B? The most obvious (but wrong) answer, that the star will shrink until it is so small that it becomes a black hole, was anathema to Eddihgton; he refused even to consider it. The only reasonable answer, he asserted, was that the star must ultimately turn cold and then support itself not by thermal pressure (that is, heat-induced pressure), but rather by the only other type of pressure known in 1925: the pressure that one finds in solid objects like rocks, a pressure due to repulsion between adjacent atoms. But such "rock pressure" was only possible, Eddington believed (incorrectly), if the star's matter has a density something like that of a rock, a few grams per cubic centimeter—10,000 times less than the present density of Sirius B.

This line of argument led to Eddington's paradox. In order to reexpand to the density of rock and thereby be able to support itself when it turns cold, Sirius B would have to do enormous work against its own gravity, and physicists did not know of any energy supply inside the star adequate for such work. "Imagine a body continually losing heat but with insufficient energy to grow cold!" Eddington wrote. "It is a curious problem and one may make many fanciful suggestions as to what actually will happen. We here leave aside the difficulty as not necessarily fatal."

Chandrasekhar had found the resolution of this 1925 paradox in R. H. Fowler's 1926 article "On Dense Matter." The resolution lay in the failure of the laws of physics that Eddington used. Those laws had to be replaced by the new quantum mechanics, which described the pressure inside Sirius B and other white dwarfs as due not to heat, but instead to a new, quantum mechanical phenomenon: the *degenerate motions of electrons,* also called *electron degeneracy.*[3]

Electron degeneracy is somewhat like human claustrophobia. When matter is squeezed to a density 10,000 times higher than that of rock, the cloud of electrons around each of its atomic nuclei gets squashed 10,000-fold. Each electron thereby gets confined to a "cell" with 10,000 times smaller volume than the one it previously was allowed to move around in. With so little